

Stat 122 (Mathematical Statistics II)

Lesson 4.1 Efficiency, Consistency, and Sufficiency

Learning Objectives

At the end of this lesson, students are expected to

1. calculate and interpret relative efficiency of estimators;
2. explain consistency of estimators; and
3. prove that a statistic is sufficient for a parameter.

Efficiency

Recall that in many problems, we are able to observe an iid sample $Y_1, Y_2, \dots, Y_n$ from a population distribution $f_Y(y, \theta)$, where θ is regarded as an unknown parameter that is to be estimated with the observed data. From the previous module, we know that a *good* estimator $\hat{\theta} = T(Y_1, Y_2, \dots, Y_n)$ has the following properties:

1. $\hat{\theta}$ is unbiased
2. $\hat{\theta}$ has the small variance

In our quest to find a good estimator for θ , we might have several *candidate estimators* to consider. For example, suppose that $\hat{\theta}_1 = T_1(Y_1, Y_2, \dots, Y_n)$ and $\hat{\theta}_2 = T_2(Y_1, Y_2, \dots, Y_n)$ are two estimators for θ . Which estimator is better? Is there a *best* estimator available? If so, how do we find it?

Definition

Suppose that $\hat{\theta}_1$ and $\hat{\theta}_2$ are unbiased estimators for θ . We call

$$RE(\hat{\theta}_1, \hat{\theta}_2) = \frac{V(\hat{\theta}_2)}{V(\hat{\theta}_1)}$$

the **relative efficiency (RE)** of $\hat{\theta}_2$ to $\hat{\theta}_1$. This is simply a ratio of the variances. If

$$\text{If } V(\hat{\theta}_1) = V(\hat{\theta}_2) \iff RE(\hat{\theta}_1, \hat{\theta}_2) = 1$$

$$\text{If } V(\hat{\theta}_1) > V(\hat{\theta}_2) \iff RE(\hat{\theta}_1, \hat{\theta}_2) < 1$$

$$\text{If } V(\hat{\theta}_1) < V(\hat{\theta}_2) \iff RE(\hat{\theta}_1, \hat{\theta}_2) > 1$$

NOTE: It only makes sense to use this measure when both $\hat{\theta}_1$ and $\hat{\theta}_2$ are unbiased estimators for θ .

Example 4.1.1

Suppose that Y_1, Y_2, Y_3 is an iid sample of $n = 3$ Poisson observations with mean θ . Consider the two candidate estimators:

$$\hat{\theta}_1 = \bar{Y}$$

$$\hat{\theta}_2 = \frac{1}{6}(Y_1 + 2Y_2 + 3Y_3)$$

1. Verify that $\hat{\theta}_1$ and $\hat{\theta}_2$ are unbiased estimators for θ . [VERIFY!]
2. Which estimator is better?

In deciding which estimator is better, we thus should compare the variances of the two estimators. Now,

$$V(\hat{\theta}_1) = V(\bar{Y}) = \frac{\theta}{3}$$

and

$$\begin{aligned} V(\hat{\theta}_2) &= V\left[\frac{1}{6}(Y_1 + 2Y_2 + 3Y_3)\right] \\ &= \left(\frac{1}{6}\right)^2 [V(Y_1) + 2^2 V(Y_2) + 3^2 V(Y_3)] \\ &= \frac{1}{36} [\theta + 4\theta + 9\theta] \\ &= \frac{1}{36} [14\theta] \\ &= \frac{7\theta}{18} \end{aligned}$$

Thus,

$$RE(\hat{\theta}_1, \hat{\theta}_2) = \frac{V(\hat{\theta}_2)}{V(\hat{\theta}_1)} = \frac{\frac{7\theta}{18}}{\frac{\theta}{3}} = \frac{7}{6} \approx 1.17$$

Therefore, $\hat{\theta}_1 = \bar{Y}$ is better or more efficient than $\hat{\theta}_2 = \frac{1}{6}(Y_1 + 2Y_2 + 3Y_3)$

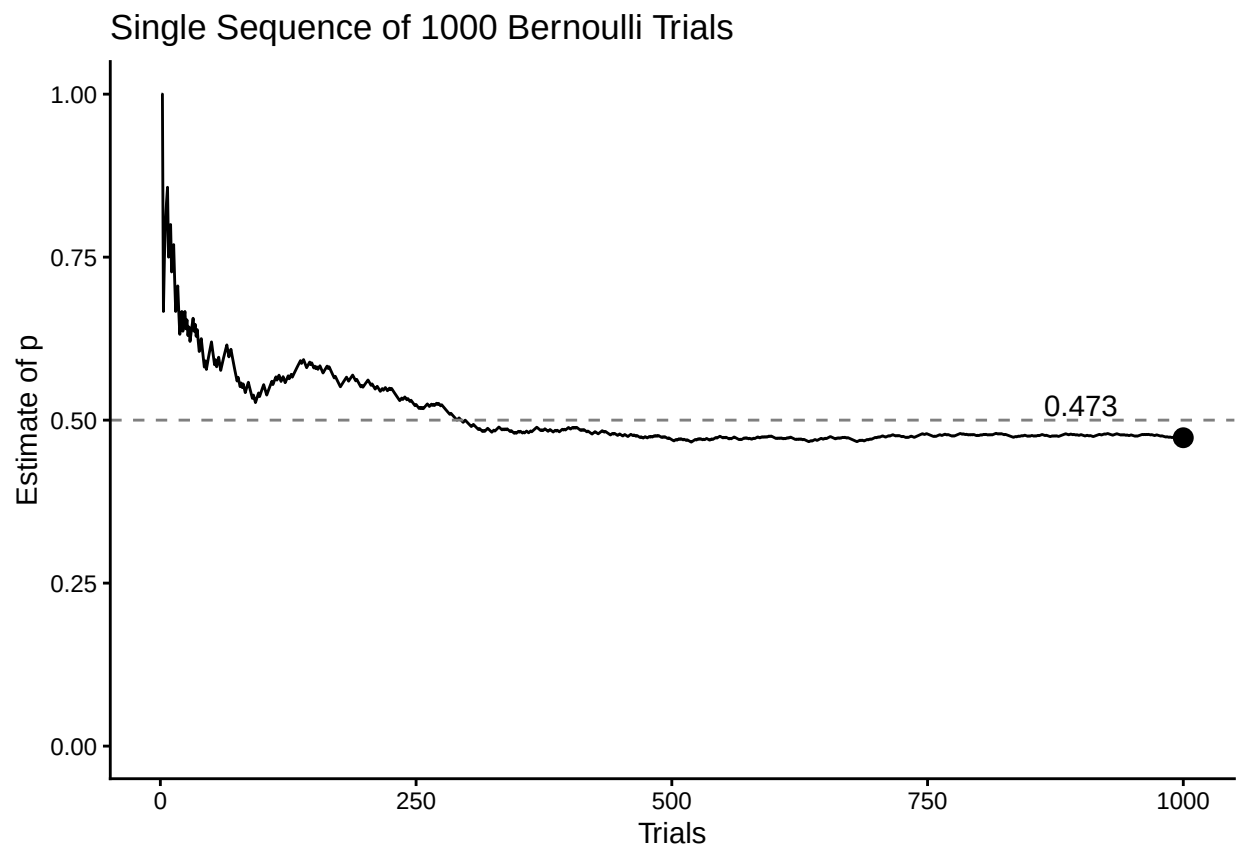
NOTE: There is not always a clear-cut winner when comparing two (or more) estimators. One estimator may perform better for certain values of θ , but be worse for other values of θ . Of course, it would be nice to have an estimator perform uniformly better than all competitors. This begs the question: Can we find the best estimator for the parameter θ ? How should we define **best**.

CONVENTION: We will define the **best** estimator as one that is **unbiased** and has the **smallest variance** among all unbiased estimators.

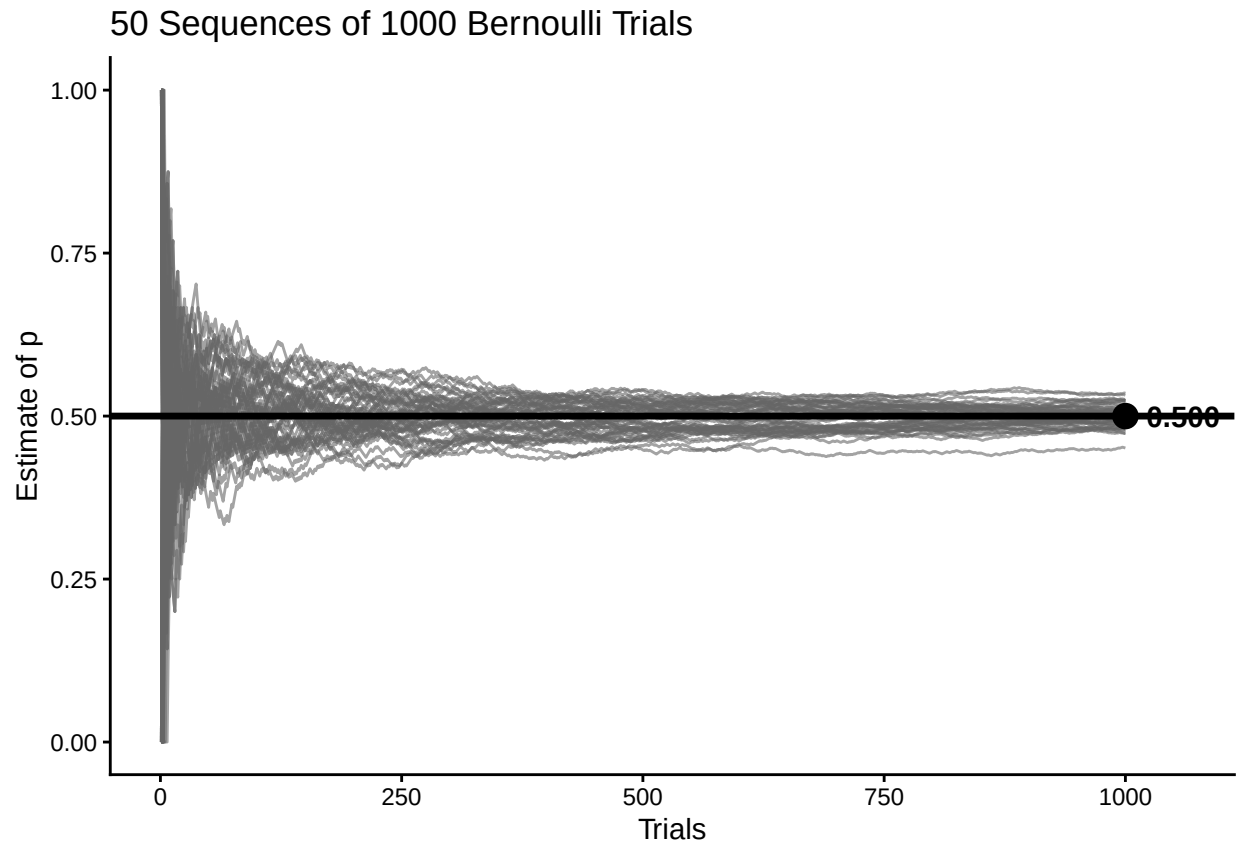
Consistency

Suppose that a coin, which has probability p of resulting in heads, is tossed n times. If the tosses are independent, then Y , the number of heads among the n tosses, has a binomial distribution. If the true value of p is unknown, the sample proportion $\hat{p} = Y/n$ is an estimator of p . What happens to this sample proportion as the number of tosses n increases? Our intuition leads us to believe that as n gets larger, $\hat{p}$ should get closer to the true value of p . That is, as the amount of information in the sample increases, our estimator should get closer to the quantity being estimated.

The figure below illustrates the values of $\hat{p}$ for a single sequence of 1000 Bernoulli trials when the true value of p is 0.5. Notice that the values of $\hat{p}$ bounce around 0.5 when the number of trials is small but approach and stay very close to $p = 0.5$ as the number of trials increases.



The next plot below shows the combined results of 50 sequences of 1000 trials. Notice that the 50 distinct sequences were not identical. This plot shows a “convergence” of sorts to the true value $p = 0.5$. This is exhibited by a wider spread of the values of the estimates for smaller numbers of trials but a much narrower spread of values of the estimates when the number of trials is larger.



Definition

The estimator $\hat{\theta}_n$ is said to be a consistent estimator of θ if, for any positive number ϵ ,

$$\lim_{n \rightarrow \infty} P(|\hat{\theta}_n - \theta| \leq \epsilon) = 1$$

or, equivalently,

$$\lim_{n \rightarrow \infty} P(|\hat{\theta}_n - \theta| > \epsilon) = 0$$

The notation $\hat{\theta}_n$ expresses that the estimator for θ is calculated by using a sample of size n . For example, $\bar{Y}_2$ is the average of two observations whereas $\bar{Y}_{100}$ is the average of the 100 observations contained in a sample of size $n = 100$. If $\hat{\theta}_n$ is an unbiased estimator, the following theorem can often be used to prove that the estimator is consistent.

Theorem

An unbiased estimator $\hat{\theta}_n$ for θ is a consistent estimator of θ if

$$\lim_{n \rightarrow \infty} V(\hat{\theta}_n) = 0$$

REMARKS:

The consistency property given in the definition and in the theorem involves a particular type of convergence of $\hat{\theta}$ to θ . For this reason, the statement “ $\hat{\theta}$ is a consistent estimator for θ ” is sometimes replaced by the equivalent statement “ $\hat{\theta}$ converges in probability to θ ”.

Example 4.1.2

Suppose that $Y_1, Y_2, \dots, Y_n$ denote a random sample from a distribution with mean μ and variance $\sigma^2 < \infty$. Show that $\bar{Y}_n = \frac{1}{n} \sum_{i=1}^n Y_i$ is a consistent estimator for μ .

SOLUTION:

Recall that $\bar{Y}_n$ is unbiased for μ , that is, $E[\bar{Y}_n] = \mu$ and $V[\bar{Y}_n] = \frac{\sigma^2}{n}$. Thus,

$$\lim_{n \rightarrow \infty} V[\bar{Y}_n] = \lim_{n \rightarrow \infty} \frac{\sigma^2}{n} = 0$$

Therefore, $\bar{Y}_n$ is a consistent estimator for μ . Equivalently, we may say that $\bar{Y}_n$ converges in probability to μ . The fact that $\bar{Y}_n$ is consistent for μ , or converges in probability to μ , is sometimes referred to as the **Law of Large Numbers**.

Sufficiency

No concept in the theory of point estimation is more important than that of sufficiency. Suppose that $Y_1, Y_2, \dots, Y_n$ is an iid sample from a population distribution $f_Y(y, \theta)$ and that the goal is to find the best estimator for θ based on $Y_1, Y_2, \dots, Y_n$. We will soon see that best estimators, if they exist, are always functions of sufficient statistics. For now, we will assume that θ is a scalar.

Definition

Suppose that $Y_1, Y_2, \dots, Y_n$ is an iid sample from a population distribution $f_Y(y, \theta)$. We call $U = g(Y_1, Y_2, \dots, Y_n)$ a **sufficient statistic** for θ if the conditional distribution of $\mathbf{Y} = (Y_1, Y_2, \dots, Y_n)$, given U , does not depend on θ .

Based on the definition, in order to show that a statistic U is sufficient, it suffices to show that the ratio

$$f_{\mathbf{Y}|U}(\mathbf{y}|u) = \frac{f_{\mathbf{Y}}(\mathbf{y}; \theta)}{f_U(u; \theta)}$$

does not depend on θ .

The Likelihood Function

Suppose that $Y_1, Y_2, \dots, Y_n$ is an iid sample from $f_Y(y, \theta)$. After we observe the data $y_1, y_2, \dots, y_n$, that is, the realizations of $Y_1, Y_2, \dots, Y_n$, we can think of the function

$$f_{\mathbf{Y}}(\mathbf{y}; \theta) = \prod_{i=1}^n f_Y(y_i; \theta)$$

in two different ways:

- (1) as the multivariate probability density/mass function of $\mathbf{Y} = (Y_1, Y_2, \dots, Y_n)$, for a fixed (but unknown) value of θ , or
- (2) as a function of θ , given the observed data $\mathbf{y} = (y_1, y_2, \dots, y_n)$.

In (1), we write

$$f_{\mathbf{Y}}(\mathbf{y}; \theta) = \prod_{i=1}^n f_Y(y_i; \theta)$$

In (2), we write

$$L(\theta|\mathbf{y}) = L(\theta|y_1, y_2, \dots, y_n) = \prod_{i=1}^n f_Y(y_i; \theta)$$

Definition

Suppose that $Y_1, Y_2, \dots, Y_n$ is an iid sample from $f_Y(y, \theta)$ and that $(y_1, y_2, \dots, y_n)$ are the n observed values. The **likelihood function** for θ is given by

$$L(\theta|\mathbf{y}) \equiv L(\theta|y_1, y_2, \dots, y_n) = \prod_{i=1}^n f_Y(y_i; \theta)$$

Example 4.1.3

Suppose that $Y_1, Y_2, \dots, Y_n$ is an iid sample of geometric random variables with parameter $0 < \theta < 1$; that is, Y_i counts the number of Bernoulli trials until the first success is observed. Recall that the $\text{geometric}(\theta)$ PMF is given by

$$f_Y(y) = \begin{cases} \theta(1 - \theta)^{y-1}, & y = 1, 2, \dots \\ 0, & \text{otherwise} \end{cases}$$

The likelihood function for θ , given the data $\mathbf{y} = (y_1, y_2, \dots, y_n)$, is

$$\begin{aligned} L(\theta|\mathbf{y}) &= \prod_{i=1}^n f_Y(y_i; \theta) \\ &= \prod_{i=1}^n \theta(1 - \theta)^{y_i-1} \\ &= \theta^n (1 - \theta)^{u-n}, \text{ where } u = \sum_{i=1}^n y_i \end{aligned}$$

Suppose we observe the number of stoplights until the first stop is required. These observations are modeled as $n = 10$ realizations from geometric distribution with parameter θ .

i	1	2	3	4	5	6	7	8	9	10
y	4	3	1	3	6	5	4	2	7	1

Using these data, we have $n = 10$ and $\sum_{i=1}^n y_i = 36$. Thus, the likelihood function $L(\theta|\mathbf{y})$ for $0 < \theta < 1$ is given by

$$\begin{aligned} L(\theta|\mathbf{y}) &= L(\theta|y_1, y_2, \dots, y_{10}) = \theta^{10} (1 - \theta)^{36-10} \\ &= \theta^{10} (1 - \theta)^{26} \end{aligned}$$

The plot of this likelihood function is shown below. In a sense, the likelihood function describes which values of θ are more consistent with the observed data $\mathbf{y}$. Looking at the plot, which values of θ are more consistent with the data?

Example 4.1.4

Suppose that $Y_1, Y_2, \dots, Y_n$ is an iid sample of Poisson observations with mean θ . Show that $U = \sum_{i=1}^n Y_i$ is a sufficient statistic for θ .

SOLUTION:

A moment-generating function argument shows that $U \sim \text{Poisson}(n\theta)$; thus, the pdf of U is given by

$$f_U(u) = \begin{cases} \frac{(n\theta)^n e^{-n\theta}}{u!}, & u = 0, 1, 2, \dots \\ 0, & \text{otherwise} \end{cases}$$

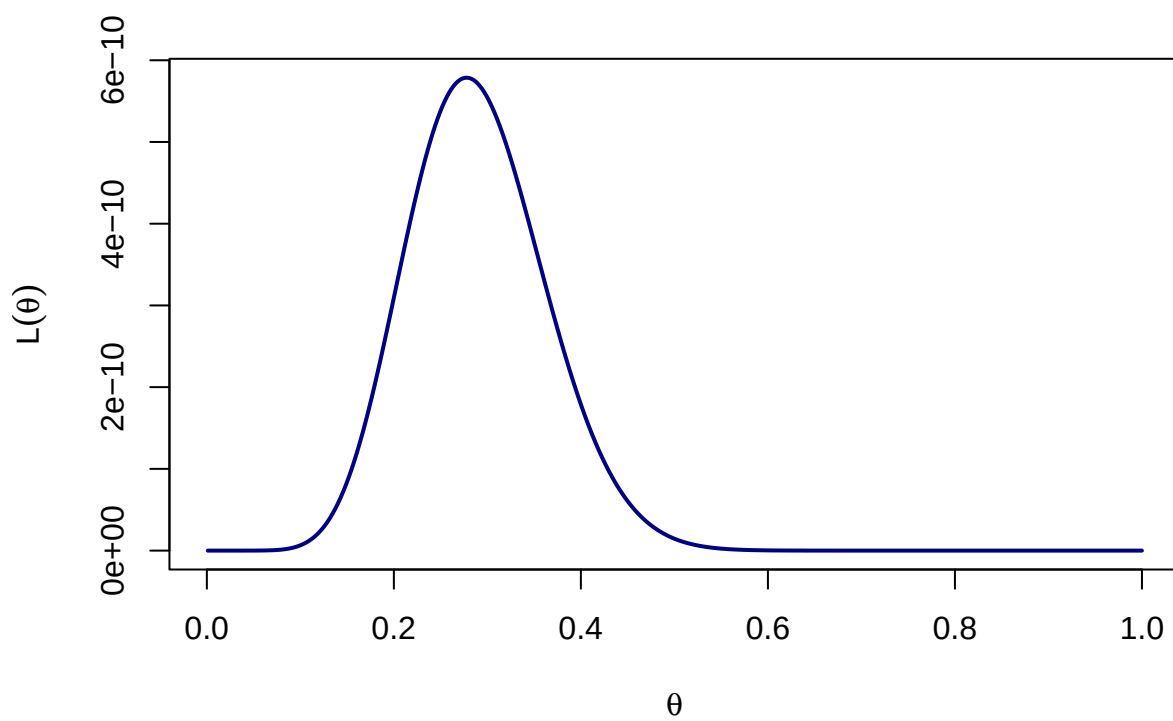


Figure 1: Likelihood function for θ

The joint distribution of the data $\mathbf{Y} = (Y_1, Y_2, \dots, Y_n)$ is the product of the marginal Poisson mass functions

$$f_{\mathbf{Y}}(\mathbf{y}; \theta) = \prod_{i=1}^n f_Y(y_i; \theta) = \prod_{i=1}^n \frac{\theta^{y_i} e^{-\theta}}{y_i!} = \frac{\theta^u e^{-n\theta}}{\prod_{i=1}^n y_i!}$$

where $u = \sum_{i=1}^n y_i$.

Therefore, the conditional distribution of $\mathbf{Y}$, given U , is equal to

$$\begin{aligned} f_{\mathbf{Y}|U}(\mathbf{y}|u) &= \frac{f_{\mathbf{Y}}(\mathbf{y}; \theta)}{f_U(u; \theta)} \\ &= \frac{\frac{\theta^u e^{-n\theta}}{\prod_{i=1}^n y_i!}}{\frac{(n\theta)^n e^{-n\theta}}{u!}} \\ &= \frac{u!}{n^u \prod_{i=1}^n y_i!} \end{aligned}$$

Since $f_{\mathbf{Y}|U}(\mathbf{y}|u)$ does not depend on the unknown parameter θ , it follows from the definition of sufficiency that $U = \sum_{i=1}^n Y_i$ is a sufficient statistic for θ .

NOTE: In a profound sense, sufficient statistics summarize all the information about the unknown parameter θ . That is, we can reduce our sample $Y_1, Y_2, \dots, Y_n$ to a sufficient statistic U and not lose any information about θ . Suppose we have two researchers. One keeps all the individual observations $Y_1, Y_2, \dots, Y_n$. The other only keeps the sum $U = \sum_{i=1}^n Y_i$ but discarded the individual observations. If both researchers wanted to estimate θ , researcher 2 has just as much information with U as researcher 1 does with the entire sample of data!

Factorization Theorem

Recall that given an iid sample $Y_1, Y_2, \dots, Y_n$ from $f_Y(y; \theta)$ in order to show that a statistic U is sufficient for θ we need to show that the conditional distribution of the data $\mathbf{Y} = (Y_1, Y_2, \dots, Y_n)$, given U , does not depend on θ . It turns out that there is an easier way to show that a statistic U is sufficient for θ .

Factorization Theorem

Suppose that $Y_1, Y_2, \dots, Y_n$ is an iid sample from $f_Y(y; \theta)$ and that U is a statistic. If the likelihood function for θ , $L(\theta; \mathbf{y})$, can be expressed as the product of two non-negative functions $g(u; \theta)$ and $h(y_1, y_2, \dots, y_n)$, where

- $g(u; \theta)$ is a function of U and θ
- $h(y_1, y_2, \dots, y_n)$ is only a function of $y_1, y_2, \dots, y_n$.

then U is a sufficient statistic for θ .

REMARK:

The Factorization Theorem makes getting sufficient statistics easy! All we have to do is be able to write the likelihood function

$$L(\theta; \mathbf{y}) = g(u; \theta) \times h(y_1, y_2, \dots, y_n)$$

for non-negative functions g and h . Now that we have the Factorization Theorem, there will rarely be a need to work directly with the conditional distribution, $f_{\mathbf{Y}|U}(\mathbf{y}|u)$, that is establish sufficiency using the definition.

Example 4.1.5

Suppose that $Y_1, Y_2, \dots, Y_n$ is an iid sample of Poisson observations with mean θ . Our goal is to show that $U = \sum_{i=1}^n Y_i$ is a sufficient statistic for θ using the Factorization Theorem.

SOLUTION:

Recall that using the definition, we shown that $U = \sum_{i=1}^n Y_i$ is sufficient by appealing to the definition of sufficiency directly.

Now, we will show that the same statistic is sufficient for θ using the Factorization Theorem. The likelihood function for θ is given by

$$\begin{aligned} L(\theta|\mathbf{y}) &= \prod_{i=1}^n f_Y(y_i; \theta) \\ &= \prod_{i=1}^n \frac{\theta^{y_i} e^{-\theta}}{y_i!} \\ &= \frac{\theta^u e^{-n\theta}}{\prod_{i=1}^n y_i!} \\ &= \underbrace{\theta^u e^{-n\theta}}_{g(u; \theta)} \times \underbrace{\left(\prod_{i=1}^n y_i! \right)^{-1}}_{h(y_1, y_2, \dots, y_n)} \end{aligned}$$

Both $g(u; \theta)$ and $h(y_1, y_2, \dots, y_n)$ are nonnegative functions. Thus, by the Factorization Theorem, $U = \sum_{i=1}^n Y_i$ is a sufficient statistic for θ .

Example 4.1.5

Suppose $Y_1, Y_2, \dots, Y_n$ is an iid sample from a Geometric distribution with parameter $0 < \theta < 1$. Show that $U = \sum_{i=1}^n Y_i$ is a sufficient statistic for θ .

SOLUTION: [A classroom exercise!]

Learning Task

INSTRUCTION: Answer the following as indicated.

1. Consider a random sample of size 3 from an exponential distribution with density function given by

$$f_Y(y) = \begin{cases} \frac{1}{\theta} e^{-\frac{y}{\theta}}, & y > 0 \\ 0, & \text{elsewhere} \end{cases}$$

- a. Show that $\hat{\theta}_1 = Y_1$, $\hat{\theta}_2 = \frac{Y_1+Y_2}{2}$, $\hat{\theta}_3 = \frac{Y_1+2Y_2}{3}$, and $\hat{\theta}_4 = \bar{Y}$ are all unbiased for θ .
 - b. Find the efficiency of $\hat{\theta}_1$ relative to $\hat{\theta}_4$, of $\hat{\theta}_2$ relative to $\hat{\theta}_4$, and of $\hat{\theta}_3$ relative to $\hat{\theta}_4$.
2. Let $Y_1, Y_2, \dots, Y_n$ be a random sample from a Uniform distribution on the interval $[0, \theta]$, where $\theta > 0$ is an unknown parameter. Show that $Y_{(n)} = \max(Y_1, Y_2, \dots, Y_n)$ is a consistent estimator of θ .
 3. Suppose $Y_1, Y_2, \dots, Y_n$ is an iid sample from a $N(0, \sigma^2)$ distribution. Show by Factorization Theorem that $U = \sum_{i=1}^n Y_i^2$ is a sufficient statistic for σ^2 .